

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ-1

DURATION: 35 MINUTES

SUMMER SEMESTER, 2022-2023

FULL MARKS: 15

Math 4641: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

1. Suppose, R and k are arbitrary numbers such that $R > 0$ and $k \neq 0$. Prove that the Newton-Raphson formula for finding the k th root of R , which is mathematically denoted as $\sqrt[k]{R}$, is as shown in Equation-1. 3
(CO2)
(PO1)

$$x_{i+1} = \frac{1}{k} \left[(k-1)x_i + \frac{R}{x_i^{k-1}} \right] \quad (1)$$

Solution:

We know, the Newton-Raphson method formula for solving $f(x) = 0$ is,

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Suppose,

$$\begin{aligned} x &= \sqrt[k]{R} \\ \Rightarrow x^k &= R \\ \Rightarrow x^k - R &= 0 \end{aligned}$$

We take $f(x) = x^k - R$ and so $f'(x) = kx^{k-1}$.

Thus,

$$\begin{aligned} x_{i+1} &= x_i - \frac{x_i^k - R}{kx_i^{k-1}} \\ &= \frac{kx_i^k - x_i^k + R}{kx_i^{k-1}} \\ &= \frac{(k-1)x_i^k + R}{kx_i^{k-1}} \\ &= \frac{1}{k} \left[\frac{(k-1)x_i^k + R}{x_i^{k-1}} \right] \\ &= \frac{1}{k} \left[(k-1)x_i + \frac{R}{x_i^{k-1}} \right] \quad \square \text{ Q.E.D.} \end{aligned}$$

Rubric:

- 1 point for using the Newton-Raphson approximation formula.
- 2 points for correct derivation.

2. a) What are significant digits? What is the relationship between the number of significant digits that are at least correct (m) in an approximation and the relative approximate error (ϵ_a)? 1
(CO2)
(PO1)

Solution:

Significant figures (also known as the significant digits, precision, or resolution) of a number in positional notation are digits in the number that are reliable and necessary to indicate the quantity of something. These are the digits that are known reliably in a reported number. The relationship between m and ϵ_a is

$$|\epsilon_a| \leq 0.5 \times 10^{2-m}\%$$

Rubric:

- 0.5 point for a correct definition. (Doesn't need to be exactly as written above.)
- 0.5 point for the correct relationship.

- b) Determine the *minimum* number of significant digits in each of these numerical values — 3
(CO4)
(PO1)
- | | | |
|--------------|------------|-------------------------------|
| (a) 69.0 | (b) 69,420 | (c) 6.94200×10^{911} |
| (d) 69.00420 | (e) 069420 | (f) 0.069420 |

Solution:

We need to determine the *minimum* number of significant digits.

- (i) 69.0 has 3 significant digits, since 0's to the right of the decimal point are significant.
- (ii) 69,420 has 4 significant digits, since trailing 0's in an integer may or may not be significant.
- (iii) 6.94200×10^{911} has 6 significant digits, since all digits of the significand/coefficient of a number's representation using Scientific Notation are significant.
- (iv) 69.00420 has 7 significant digits, since 0's between two significant non-zero digits are significant and trailing 0's in a number with the decimal point are significant.
- (v) 069420 has 4 significant digits, since leading 0's are not significant and trailing 0's in an integer may or may not be significant.
- (vi) 0.069420 has 5 significant digits, since leading 0's are not significant and trailing 0's in a number with the decimal point are significant.

Rubric:

- 0.5 point for each correct answer. (The reasons are not required.)

3. “If a function $f(x)$ does **not** change its sign between two points x_l and x_u , then there may or may not be any root of the equation $f(x) = 0$ between them.” – Do you agree? Justify your answer. 3
(CO3)
(PO2)

Solution:

Yes, I agree.

There are 4 possible cases for which we get $f(x_l)f(x_u) > 0$. If we examine the 1st and 2nd cases (as shown in Figure-1 and Figure-2 respectively), we observe that there can be multiple roots between x_l and x_u .

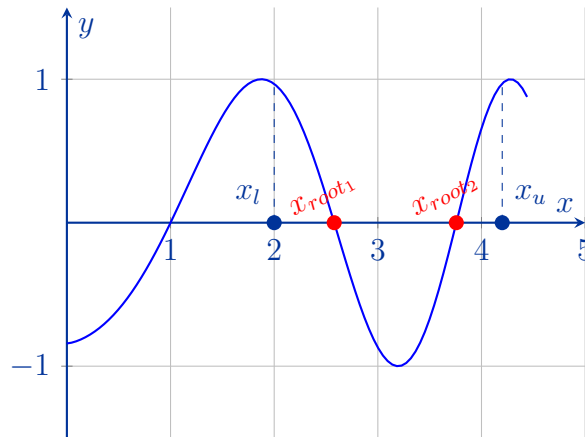


Figure 1: The 1st case (both $f(x_l)$ and $f(x_u)$ are positive, and roots exist).

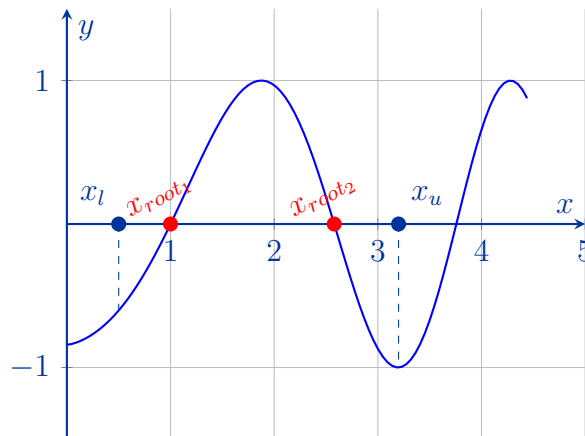


Figure 2: The 2nd case (both $f(x_l)$ and $f(x_u)$ are negative, and roots exist).

If we examine the 3rd and 4th cases (as shown in Figure-3 and Figure-4 respectively), we observe that there are no roots between x_l and x_u .

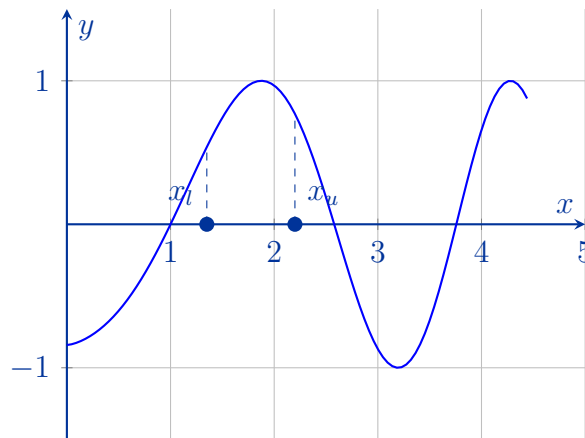


Figure 3: The 3rd case (both $f(x_l)$ and $f(x_u)$ are positive, but roots don't exist).

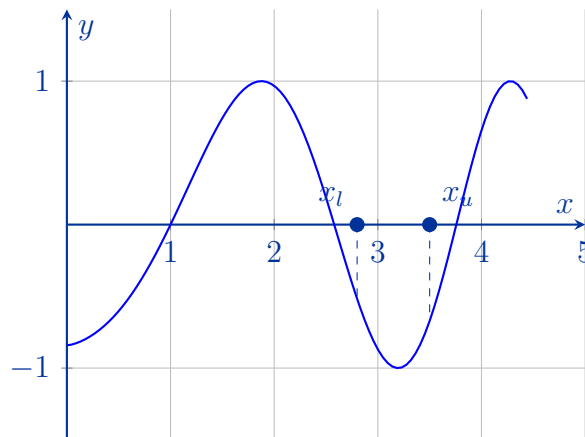


Figure 4: The 4th case (both $f(x_l)$ and $f(x_u)$ are negative, but roots don't exist).

Rubric:

- 1 point for the correct verdict.
- 0.5 point for each possible case.

4. The *Plastic Ratio*, often denoted by the Greek letter ρ (rho), is the ratio involving three quantities a , b , and c , such that $a > b > c > 0$, if they satisfy the equality $\frac{a}{b} = \frac{b+c}{a} = \frac{b}{c}$. The constant ρ is equal to the unique real root of the cubic equation $x^3 = x + 1$. The numerical value of that root is,

$$x_{root} = \rho = \sqrt[3]{\frac{9 + \sqrt{69}}{18}} + \sqrt[3]{\frac{9 - \sqrt{69}}{18}} = 1.324717957 \dots$$

Apply the *Regula Falsi* method, also known as the False Position method, to estimate the value of the plastic ratio ρ . The initial upper and lower guesses are $x_l = -1$ and $x_u = 2$ respectively. Conduct 4 iterations, and find the relative approximate error ($|\epsilon_a|\%$) and the number of significant digits that are at least correct (m), at the end of each iteration. Fill out the following table after performing the necessary calculations.

Iteration	x_l	x_u	x_r	$ \epsilon_a \%$	m	$f(x_r)$
1	-1	2		—	—	
2						
3						
4						

Solution:

After performing the series of calculations, we end up with the following table.

Iteration	x_l	x_u	x_r	$ \epsilon_a \%$	m	$f(x_r)$
1	-1	2	-0.5	—	—	-0.625
2	-0.5	2	-0.2222	125	0	-0.7887
3	-0.2222	2	0.0805	376.0248	0	-1.08
4	0.0805	2	0.4215	80.9015	0	-1.3466

Rubric:

- 0.25 point for each correct answer.